

# Mantu

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## Skills

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**Languages:** JavaScript, TypeScript, Swift, Java

**Technologies:** React Native, Expo, React, Next.js, Redux, Zustand, React Query, Jest, Firebase, GitHub, CI/CD

**Tools:** Xcode, Android Studio, VSCode, Cursor, Postman, Antigravity

## Experience

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**Sr. React Native Developer & Team Lead**, iTechnolabs Pvt. Ltd. – Mohali, Punjab Aug 2022 – present

- Led and mentored a team of 12+ developers, enforcing TypeScript standards that eliminated runtime errors and boosted team productivity by 25%.
- Integrated Google's MediaPipe PoseLandmark model for real-time human pose estimation in a fitness app, achieving 95% pose accuracy across 33 body landmarks for workout tracking.
- Architected a Turbo Module for human pose detection supporting React Native's New Architecture, and extended Expo compatibility via a custom config plugin.
- Created a production-ready React Native boilerplate (Expo SDK 54) featuring TypeScript, Zustand, TanStack Query, i18n, ESLint, Prettier, and Husky pre-commit hooks.
- Engineered a high-performance POS system (React Native + Next.js), boosting operational efficiency by 30% and user satisfaction by 20%.
- Accelerated app performance by 20% through the implementation of a new bridgeless architecture and cut project setup time by 30% by migrating to Expo.
- Managed end-to-end client communication and delivery for key accounts, achieving a 95% client satisfaction rate through transparent reporting.
- Resolved critical Google Maps integration bugs, cutting client API costs by 40% and preventing billing leakages.

**Sr. React Native Developer**, Suffescom Solutions – Mohali, Punjab Oct 2021 – Aug 2022

- Directed and coached a team of 7 developers, establishing React Native coding standards that enhanced collaboration and code quality.
- Created an Android native module enabling background tasks with 99.9% reliability.
- Built a food delivery app serving 1,000+ users, implementing real-time tracking and payments via WebSockets and Maps.
- Enhanced app security by resolving subscription loopholes, ensuring 100% compliance with single-user policies.

**Junior iOS Developer & React Native Developer**, Immanent Solutions – Mohali, Punjab Sept 2018 – Oct 2021

- Engineered 4 scalable mobile apps for event, education, social, and crypto sectors, serving 1,000+ users via iOS and React Native.
- Achieved 100% security compliance by implementing HMAC-based API authentication to mitigate unauthorized access risks.
- Shortened development time by 40% by migrating native iOS codebases to React Native.

## Open Source

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### expo-check-installed-apps

Expo config plugin to detect installed apps on Android and iOS (256.8K+ downloads)

- Created an Expo module enabling developers to query device-installed apps for feature gating.
- Supports Expo SDK 51+ with automatic native code injection via config plugin.

### **react-native-mediapipe-posedetection**

Real-time pose detection for React Native using MediaPipe BlazePose with GPU acceleration (8.7K+ downloads)

- Crafted a high-performance native module supporting 33 pose landmarks with ~15 FPS real-time detection.
- Implemented GPU-accelerated inference for iOS and Android using Vision Camera frame processing.

## **Projects**

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### **Focusbear - ADHD**

Productivity app helping users block distractions and build healthy habits.

- Built native modules for Android (distraction-blocking dialog) and iOS (Screen Time API), achieving a 50% decrease in user distractions.
- Formulated a scheduled app-blocking feature, elevating user focus consistency by 40% during routine intervals.
- Deployed a log file system to track errors and crashes, boosting error resolution efficiency by 50%.
- Introduced offline API caching, ensuring 99.9% uptime and an uninterrupted user experience.

### **Elevyn - AI Fitness**

AI-powered fitness app with personalized workouts, meal planning, and calorie tracking.

- Integrated Google's MediaPipe PoseLandmarker for on-device human pose estimation, enabling real-time workout form analysis across 33 body landmarks.
- Built an AI-driven engine generating personalized workout and meal plans based on user body metrics, increasing user engagement by 35%.
- Implemented camera-based meal scanning to auto-detect macros and calories, streamlining food logging and boosting daily tracking consistency by 40%.

### **JOBR - POS**

Point of Sale system for service management.

- Crafted a custom calendar module for service scheduling, enhancing appointment management efficiency by 40%.
- Integrated Apple's Tap to Pay for secure transactions, increasing payment versatility by 50%.
- Created a wallet module for JOBR coins, boosting in-app transaction engagement and user retention by 30%.
- Built a web app using Next.js, ensuring consistent cross-platform functionality and refining UX by 40%.

### **Mealed - Food Ordering**

On-demand food ordering and delivery platform.

- Integrated React Native Maps for real-time driver tracking, enhancing order capabilities by 50%.
- Added Stripe for card payments and created a wallet module, enhancing transaction convenience by 40%.
- Implemented real-time push notifications using Notifee, boosting user engagement for order updates by 25%.
- Incorporated Agora for in-app calling, elevating communication efficiency by 35%.

### **Ride2Duty**

Corporate mobility platform streamlining employee commutes via automated scheduling.

- Architected a corporate commute platform using Expo, enabling automated scheduling and recurring ride management for seamless employee transportation.
- Designed a resilient background location service that bypasses Android battery optimizations (Doze mode), ensuring 100% real-time tracking reliability.
- Created stutter-free map animations by implementing custom coordinate interpolation for driver markers, ensuring smooth real-time location updates.
- Established critical permission handling flows to secure 'Always Allow' access, resolving background execution failures across diverse Android OEM devices.

## **Education**

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